MetaMask Practicals – Detailed Theory Notes

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Practical 1: Create a MetaMask Account

Aim: To install MetaMask and create a cryptocurrency wallet account for interacting with blockchain networks.

Theory: MetaMask is a digital cryptocurrency wallet that allows users to create and manage blockchain accounts. It is available as a browser extension and a mobile application. It acts as an interface between the user and supported blockchain networks, allowing users to view assets, send and receive tokens, and connect to decentralised applications (dApps).

A MetaMask wallet is generally described as a **non-custodial wallet**, meaning the user controls the wallet credentials rather than relying on a central service to authorize transactions. When a wallet is created, cryptographic keys are generated. The **public wallet address** can be shared with others to receive cryptocurrency. The **private key** and **Secret Recovery Phrase** provide control over the wallet and must be kept confidential. Anyone who obtains these secret credentials may be able to access the wallet.

MetaMask does not hold cryptocurrency like a bank account. Cryptocurrency balances and transaction records are maintained on the blockchain; MetaMask provides a convenient way to access and manage them.

Requirements: A computer or smartphone, internet connection, supported browser or MetaMask mobile app, and a secure password.

Procedure

- Open the official MetaMask website at https://metamask.io/ and choose the appropriate browser extension or mobile application.
- Install MetaMask and open the application from the browser toolbar or device.
- Choose the option to create a new wallet and follow the on-screen instructions.
- Create a strong password to protect access to MetaMask on that device.
- Complete the wallet setup and securely back up the Secret Recovery Phrase when prompted. Never share it with another person or enter it on an unknown website.
- Open the account details to view and copy the public wallet address. This address can be shared to receive supported cryptocurrency.
- Check the selected blockchain network before receiving or sending assets, because addresses and assets must be used on compatible networks.

Important terms:

Term	Description
Wallet	A tool that manages cryptographic keys and enables users to interact with blockchain assets.
Public address	An account identifier that can be shared to receive cryptocurrency.
Private key	A secret cryptographic key used to authorize control of an account. It must never be shared.
Secret Recovery Phrase	A set of words used to recover the wallet. Anyone who has it may gain control of the wallet.
dApp	A decentralised application that interacts with a blockchain through a wallet.

Result: MetaMask was installed and a new wallet account was created successfully. The public wallet address was obtained.

Conclusion: Learned the purpose of MetaMask, how to create a wallet account, and the difference between public and secret wallet credentials.

Practical 2: Create a MetaMask Wallet and Perform Transactions

Aim: To use MetaMask to send and receive cryptocurrency and verify transactions on a blockchain network.

Theory: MetaMask allows users to interact with blockchain networks by sending and receiving cryptocurrency. A transaction generally contains the sender address, receiver address, amount, selected network and a network fee. Before submitting a transaction, the user reviews the details in MetaMask and confirms it. The blockchain network then validates and records the transaction.

Every transaction on Ethereum requires computational resources. The network fee, commonly called the **gas fee**, compensates the network for processing the transaction. The fee depends on the amount of gas used and the effective gas price. The amount sent to the receiver and the gas fee are separate values.

After submission, the transaction receives a unique **transaction hash (Tx Hash)**. This hash can be entered into a blockchain explorer to view details such as transaction status, sender, receiver, transferred value, gas used and transaction fee. For educational practice, users can select the **Sepolia test network** and use test ETH. Test ETH is intended for testing and does not have the same monetary value as real ETH.

Requirements: MetaMask wallet, internet connection, Ethereum Sepolia test network, test ETH, a second wallet address for receiving, and Sepolia Etherscan.

Procedure

- Open MetaMask and unlock the wallet using the device password.
- Enable test networks if necessary, then select **Ethereum Sepolia** from the network list.
- Obtain Sepolia test ETH from a testnet faucet if the wallet does not have enough test ETH.
- Create or select a second MetaMask account to act as the receiver, and copy its public wallet address.
- Switch back to the sending account and select **Send**.
- Paste the receiver's public address carefully and select the correct asset, such as SepoliaETH.
- Enter the amount to transfer, for example 0.001 ETH, and continue.
- Review the receiver address, transfer amount and estimated network fee. Confirm that the selected network is Sepolia.
- Confirm the transaction in MetaMask and wait for the transaction to be processed.
- Open the wallet activity, copy the transaction hash and visit Sepolia Etherscan.
- Paste the transaction hash into the explorer and inspect the transaction status, sender, receiver, value, gas used and transaction fee.
- Switch to the receiving account and verify that the test ETH has arrived.

Important terms

Term	Description
Transaction	An instruction submitted to a blockchain, such as transferring cryptocurrency.
Transaction hash	A unique identifier used to locate and verify a transaction on a blockchain explorer.
Gas	A measurement of the computational work required to execute an Ethereum transaction.
Gas fee	The network fee for processing a transaction. It is calculated using gas used and effective gas price.
Test network	A blockchain network used for development and practice without using real-value cryptocurrency.
Blockchain explorer	A website used to inspect publicly available blockchain transactions and addresses.

Gas fee formula: Transaction Fee = Gas Used × Effective Gas Price. The displayed fee is paid in the network's native currency. The transfer amount is not the gas fee.

Result: Successfully sent test ETH from one MetaMask account to another and verified the transaction details using Sepolia Etherscan.

Conclusion: Learned how to perform a cryptocurrency transaction using MetaMask, understand network fees and verify a transaction through a blockchain explorer.

Precautions

- Download MetaMask only from its official website or verified app store listing.
- Never share the Secret Recovery Phrase or private key with anyone.
- Always verify the receiver address and selected blockchain network before confirming a transaction.
- Use test ETH on Sepolia for practice; do not use real ETH or the Ethereum Mainnet for this exercise.
- Check the estimated network fee before confirming a transaction.

References

1. MetaMask official website: https://metamask.io/
2. MetaMask Support – Gas fees: https://support.metamask.io/more-web3/learn/user-guide-gas
3. Ethereum – Gas and fees: https://ethereum.org/gas/
4. Sepolia Etherscan: https://sepolia.etherscan.io/